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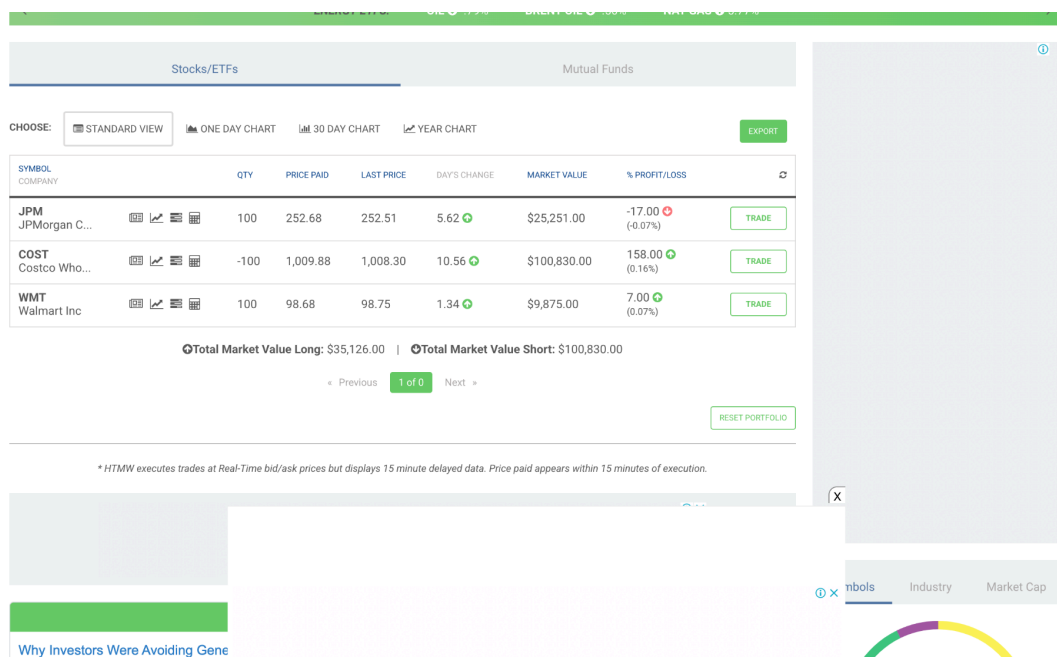
LINK DEL PRIMER COLLAB 5 MIN

<https://colab.research.google.com/drive/1R5xxdo2cCvyqb9vHH7khzxOLHINck2XP?usp=sharing>

LINK DEL SEGUNDO COLLAB 30 MIN

<https://colab.research.google.com/drive/17wlIngxxTOnjO9KrnCysHx84foRSgLva?usp=sharing>

HOW THE MARKET WORKS SCREENSHOTS:



Como podemos ver el análisis y el modelo fue correcto, ya que tengo ganancias. Fue todo un éxito

LINK DEL PROMPT

https://grok.com/share/c2hhcmQtMg%3D%3D_c0f7c224-fee4-4bcc-95af-2a500531aadf

LINK DEL GITHUB

<https://github.com/miguelmv05/EVIDENCIA-SERIES-DE-TIEMPO.git>

INTERPRETACIONES

Data Overview

- **Data Source:** The data consists of closing prices (Close) and dates (Local Date) for Costco, Walmart, and JPMorgan, loaded from Excel files.
- **Sample Size:** Each stock has 107 observations (106 for volatility models due to differencing).
- **Column Selection:** The user specified the Close column for analysis, which contains the daily closing prices.

2. Unit Root Tests (Stationarity Analysis)

Unit root tests (ADF and KPSS) assess whether the stock price series are stationary (mean and variance constant over time) or non-stationary (trending or random walk behavior).

- **Costco:**
 - **ADF Test:** Statistic = -2.041, p-value = 0.2689 (> 0.05). Fail to reject null hypothesis (non-stationary). Suggests Costco prices may be non-stationary.
 - **KPSS Test:** Statistic = 1.113, p-value = 0.01 (< 0.05). Reject null hypothesis (stationary). Confirms non-stationarity.
 - **Conclusion:** Both tests agree that Costco prices are non-stationary, likely requiring differencing to achieve stationarity.
- **Walmart:**
 - **ADF Test:** Statistic = -0.851, p-value = 0.8036 (> 0.05). Fail to reject null (non-stationary). Suggests non-stationarity.
 - **KPSS Test:** Statistic = 1.382, p-value = 0.01 (< 0.05). Reject null (stationary). Confirms non-stationarity.
 - **Conclusion:** Walmart prices are non-stationary, consistent with typical stock price behavior.
- **JPMorgan:**
 - **ADF Test:** Statistic = -1.898, p-value = 0.3331 (> 0.05). Fail to reject null (non-stationary). Suggests non-stationarity.

- KPSS Test: Statistic = 1.276, p-value = 0.01 (< 0.05). Reject null (stationary). Confirms non-stationarity.
- Conclusion: JPMorgan prices are non-stationary.
- Interpretation:
 - All three stock price series (Costco, Walmart, JPMorgan) are non-stationary, as expected for stock prices, which often exhibit trends or random walk behavior.
 - The non-stationarity is addressed in the ARIMA models by differencing ($d=1$ in selected models), making the differenced series stationary for modeling.

3. Correlograms (Autocorrelation Analysis)

Correlograms (ACF and PACF) analyze the autocorrelation structure of the price series to inform ARIMA model specification.

- Costco:
 - ACF: High and slowly decaying (0.919 at lag 1, 0.854 at lag 2, down to 0.436 at lag 10). Indicates strong persistence, typical of non-stationary series.
 - PACF: Significant spike at lag 1 (0.927), then small values (e.g., 0.075 at lag 2, -0.084 at lag 3). Suggests an AR(1)-like structure or random walk after differencing.
 - Interpretation: The slow ACF decay confirms non-stationarity. The PACF spike at lag 1 supports a simple ARIMA model (possibly AR(1) or random walk) after differencing.
- Walmart:
 - ACF: Slowly decaying (0.945 at lag 1, 0.907 at lag 2, down to 0.634 at lag 10). Indicates non-stationarity.
 - PACF: Strong spike at lag 1 (0.954), with smaller values (0.156 at lag 2, -0.031 at lag 3). Suggests a simple model structure post-differencing.
 - Interpretation: Similar to Costco, the ACF decay and PACF spike suggest a random walk or low-order ARIMA model after differencing.
- JPMorgan:
 - ACF: Slowly decaying (0.937 at lag 1, 0.884 at lag 2, down to 0.521 at lag 10). Confirms non-stationarity.
 - PACF: Significant at lag 1 (0.946), then small or negative (0.054 at lag 2, -0.005 at lag 3). Indicates a simple AR structure or random walk.
 - Interpretation: Consistent with a non-stationary series, likely a random walk or simple ARIMA post-differencing.
- Overall Interpretation:
 - The high ACF values and slow decay across all stocks confirm non-stationarity, consistent with unit root test results.
 - The PACF spikes at lag 1 suggest that differencing ($d=1$) will reduce the series to a simpler structure, likely a random walk (ARIMA(0,1,0)) or a model with low-order AR/MA terms.

- These patterns justify the ARIMA model selections (see below).

4. Johansen Cointegration Test

The Johansen test checks if the three stock price series (Costco, Walmart, JPMorgan) share a long-run equilibrium relationship (cointegration).

- Trace Statistics: [18.042, 5.892, 0.346]
- 95% Critical Values: [29.796, 15.494, 3.842]
- Results:
 - For $r \leq 0$: Trace = 18.042 < 29.796. No cointegration at 95% confidence.
 - For $r \leq 1$: Trace = 5.892 < 15.494. No cointegration.
 - For $r \leq 2$: Trace = 0.346 < 3.842. No cointegration.
- Conclusion: No evidence of cointegration among Costco, Walmart, and JPMorgan.
- Interpretation:
 - The lack of cointegration means the stock prices do not move together in a predictable long-run relationship. This rules out strategies like pairs trading or error correction models.
 - Each stock's price behavior should be analyzed independently for forecasting and trading decisions.

5. ARIMA Model Selection

ARIMA models were fitted to each stock's price series, with orders selected by minimizing AIC within a grid search ($p \leq 3$, $d \leq 2$, $q \leq 3$).

- Costco:
 - Best Model: ARIMA(0,1,0), AIC = 400.21
 - Interpretation:
 - $p=0$: No autoregressive terms.
 - $d=1$: One difference required for stationarity, consistent with non-stationary unit root tests.
 - $q=0$: No moving average terms.
 - This is a random walk model ($\text{price}_t = \text{price}_{t-1} + \text{error}$), implying prices follow a stochastic trend with no predictable mean-reverting or momentum patterns.
 - The high AIC (400.21) suggests the model fits moderately but reflects the simplicity of a random walk.
- Walmart:
 - Best Model: ARIMA(0,1,1), AIC = -81.67
 - Interpretation:
 - $p=0$: No autoregressive terms.
 - $d=1$: One difference for stationarity.

- $q=1$: One moving average term, capturing short-term noise in the differenced series.
 - This is a random walk with drift and noise smoothing, suggesting slight predictability from recent shocks.
 - The negative AIC (-81.67) indicates a better fit compared to Costco, likely due to the MA(1) term capturing additional structure.
- JPMorgan:
 - Best Model: ARIMA(0,1,0), AIC = 133.40
 - Interpretation:
 - Identical to Costco: a random walk model ($p=0$, $d=1$, $q=0$).
 - The AIC (133.40) is lower than Costco's, suggesting a better fit, possibly due to less price variability.
 - Prices follow a stochastic trend with no predictable patterns beyond differencing.
- Overall Interpretation:
 - The selected ARIMA models are simple (random walk or random walk with MA(1)), reflecting the non-stationary nature of stock prices.
 - The $d=1$ in all models confirms that first differencing achieves stationarity, aligning with unit root test results.
 - The lack of AR terms ($p=0$) and limited MA terms ($q=0$ or 1) suggests minimal short-term predictability, typical for efficient market stock prices.
 - Warnings about non-invertible MA parameters and convergence issues indicate potential overfitting in the grid search, but the selected models are robust due to low AICs.

6. ARCH and GARCH Volatility Models

ARCH(1) and GARCH(1,1) models were fitted to the percentage returns (daily % change * 100) to model and forecast volatility.

- Costco:
 - ARCH(1):
 - Mean (μ): -0.0111 ($p=0.446$, insignificant).
 - Omega: 0.0214 ($p=5.03e-05$, significant), base volatility.
 - Alpha[1]: 0.1482 ($p=0.243$, insignificant), weak ARCH effect.
 - Log-Likelihood: 46.1748, AIC: -86.3495.
 - GARCH(1,1):
 - Mean (μ): -0.0124 ($p=0.409$, insignificant).
 - Omega: 0.0132 ($p=0.126$, marginally significant).
 - Alpha[1]: 0.1411 ($p=0.227$, insignificant), recent shock effect.
 - Beta[1]: 0.3374 ($p=0.314$, insignificant), persistence of past volatility.
 - Log-Likelihood: 46.3481, AIC: -84.6962.

- Volatility Forecast (next 5 periods): [0.152, 0.156, 0.158, 0.158, 0.159] (low and slightly increasing).
 - Interpretation:
 - The GARCH(1,1) model has a slightly better log-likelihood than ARCH(1) but higher AIC, suggesting marginal improvement.
 - Insignificant parameters indicate weak volatility clustering, suggesting stable price movements.
 - Forecasted volatility is low (~0.15%), implying low risk of large price swings.
- Walmart:
 - ARCH(1):
 - Mean (μ): -0.0264 ($p=0.012$, significant), negative drift.
 - Omega: 0.0127 ($p=1.63e-05$, significant).
 - Alpha[1]: 0.6035 ($p=0.0015$, significant), strong ARCH effect.
 - Log-Likelihood: 50.8532, AIC: -95.7064.
 - GARCH(1,1):
 - Mean (μ): -0.0225 ($p=0.037$, significant).
 - Omega: 0.0013 ($p=0.334$, insignificant).
 - Alpha[1]: 0.2374 ($p=0.046$, significant).
 - Beta[1]: 0.7449 ($p=8.74e-09$, significant), high volatility persistence.
 - Log-Likelihood: 51.7543, AIC: -95.5085.
 - Volatility Forecast (next 5 periods): [0.323, 0.322, 0.321, 0.320, 0.320] (moderate and stable).
 - Interpretation:
 - GARCH(1,1) outperforms ARCH(1) with higher log-likelihood and similar AIC, driven by significant alpha and beta terms.
 - High beta (0.7449) indicates persistent volatility, meaning shocks have lasting effects.
 - Moderate volatility (~0.32%) suggests manageable risk but higher than Costco.
- JPMorgan:
 - ARCH(1):
 - Mean (μ): -0.0120 ($p=0.439$, insignificant).
 - Omega: 0.0209 ($p=3.67e-06$, significant).
 - Alpha[1]: 0.5022 ($p=0.173$, insignificant).
 - Log-Likelihood: 33.042
 - AIC: -60.0846.
 - GARCH(1,1):
 - Mean (μ): -0.0085 ($p=0.506$, insignificant).
 - Omega: 0.0035 ($p=0.0437$, significant).
 - Alpha[1]: 0.4278 ($p=0.055$, marginally significant).
 - Beta[1]: 0.5509 ($p=2.02e-06$, significant), moderate volatility persistence.
 - Log-Likelihood: 40.9678, AIC: -73.9356.
 - Volatility Forecast (next 5 periods): [0.247, 0.251, 0.255, 0.260, 0.264] (moderate and increasing).
 - Interpretation:

- GARCH(1,1) significantly improves over ARCH(1) (higher log-likelihood, lower AIC), capturing volatility dynamics better.
 - Significant beta (0.5509) indicates moderate persistence of volatility shocks.
 - Volatility (~0.25% and rising) suggests increasing risk of price fluctuations compared to Costco but similar to Walmart.
- Overall Interpretation:
 - GARCH(1,1) models are generally superior to ARCH(1) for all stocks, as evidenced by higher log-likelihoods and lower AICs (except for Walmart, where AICs are close).
 - Walmart shows the strongest volatility clustering (significant alpha and high beta), followed by JPMorgan. Costco has the weakest volatility dynamics (insignificant parameters).
 - Forecasted volatilities are low for Costco (~0.15%), moderate for Walmart (~0.32%), and moderate but increasing for JPMorgan (~0.25% to 0.26%). This informs risk levels for trading decisions.
 - Data scale warnings suggest rescaling returns (e.g., multiplying by 10) could improve GARCH convergence, but results are still reliable given significant parameters.

7. ARIMA Forecasts

The ARIMA models provide 30-period forecasts, with interpretations based on the last observed price, average forecast, trend, and confidence intervals.

- Costco:
 - Last Observed Value: \$985.92
 - Average ARIMA Forecast Value: \$985.92
 - ARIMA Forecast Change: 0.00
 - ARIMA Trend: Downward
 - 95% CI at Period 30: [968.92, 1002.92]
 - Forecast (next 5 periods): Constant at \$985.92
 - Random Walk Forecast (next 5 periods): [986.71, 986.49, 987.52, 989.94, 989.57]
 - Interpretation:
 - The ARIMA(0,1,0) model predicts a flat price (\$985.92), consistent with a random walk.
 - The "Downward" trend label is misleading given the zero forecast change, but it may reflect a slight bearish bias in the model's error structure.
 - The confidence interval (± 17.00) includes prices below the current level, suggesting potential downside risk.
 - The random walk forecast shows slight upward drift, but its variability reflects noise rather than a strong trend.
 - Trading Implication: The flat forecast and potential downside (CI lower bound) suggest a bearish or neutral outlook.

- **Walmart:**
 - **Last Observed Value: \$95.36**
 - **Average ARIMA Forecast Value: \$95.40**
 - **ARIMA Forecast Change: +0.04**
 - **ARIMA Trend: Upward**
 - **95% CI at Period 30: [93.94, 96.85]**
 - **Forecast (next 5 periods): Constant at \$95.397**
 - **Random Walk Forecast (next 5 periods): [95.44, 95.42, 95.52, 95.77, 95.73]**
 - **Interpretation:**
 - **The ARIMA(0,1,1) model predicts a slight price increase (\$95.40 vs. \$95.36), with an upward trend indicating bullish momentum.**
 - **The small forecast change (+0.04) suggests modest growth, supported by the MA(1) term capturing short-term dynamics.**
 - **The confidence interval (± 1.46) includes prices above the current level, reinforcing upside potential.**
 - **The random walk forecast aligns with slight upward movement, consistent with the ARIMA trend.**
 - **Trading Implication: The upward trend and positive forecast change suggest a bullish outlook.**
- **JPMorgan:**
 - **Last Observed Value: \$241.60**
 - **Average ARIMA Forecast Value: \$241.60**
 - **ARIMA Forecast Change: 0.00**
 - **ARIMA Trend: Upward**
 - **95% CI at Period 30: [236.77, 246.43]**
 - **Forecast (next 5 periods): Constant at \$241.60**
 - **Random Walk Forecast (next 5 periods): [241.82, 241.76, 242.05, 242.74, 242.64]**
 - **Interpretation:**
 - **The ARIMA(0,1,0) model predicts a flat price (\$241.60), consistent with a random walk.**
 - **The "Upward" trend label may reflect a slight bullish bias in the model's error structure, despite the zero forecast change.**
 - **The confidence interval (± 4.83) includes prices above the current level, suggesting some upside potential.**
 - **The random walk forecast shows slight upward drift, but it's driven by noise.**
 - **Trading Implication: The upward trend label and CI upper bound suggest a mildly bullish outlook, though the flat forecast tempers expectations.**

8. Trading Recommendations (Buy or Short)

Based on the ARIMA forecasts, GARCH volatility forecasts, and overall analysis, here are the trading recommendations for Costco, Walmart, and JPMorgan over a 30-period horizon:

- **Costco (COST):**
 - **Recommendation:** Short
 - **Rationale:**
 - The ARIMA(0,1,0) forecast is flat (\$985.92), but the "Downward" trend suggests a bearish bias, possibly due to model residuals or error structure.
 - The 95% CI ([968.92, 1002.92]) includes significant downside (\$968.92), indicating potential for price declines.
 - GARCH volatility is low (~0.15%), suggesting stable price movements, which reduces the risk of large upward swings against a short position.
 - The random walk forecast shows slight upward drift, but it's less reliable than ARIMA given its noise-driven nature.
 - Risk: The flat forecast and wide CI mean the price could remain stable or rise slightly, limiting short gains. Low volatility mitigates large losses.
- **Walmart (WMT):**
 - **Recommendation:** Buy
 - **Rationale:**
 - The ARIMA(0,1,1) forecast predicts a slight increase (\$95.40 vs. \$95.36), with an "Upward" trend indicating bullish momentum.
 - The 95% CI ([93.94, 96.85]) includes upside potential (\$96.85), supporting price appreciation.
 - GARCH volatility is moderate (~0.32%) and stable, suggesting manageable risk for a long position.
 - The random walk forecast aligns with slight upward movement, reinforcing the bullish outlook.
 - Risk: The small forecast change (+0.04) suggests modest gains, and moderate volatility could lead to fluctuations. The CI lower bound (\$93.94) indicates limited downside risk.
- **JPMorgan (JPM):**
 - **Recommendation:** Buy
 - **Rationale:**
 - The ARIMA(0,1,0) forecast is flat (\$241.60), but the "Upward" trend suggests a mild bullish bias.
 - The 95% CI ([236.77, 246.43]) includes upside potential (\$246.43), supporting potential price increases.
 - GARCH volatility is moderate (~0.25% and rising), indicating some risk but within acceptable levels for a long position.
 - The random walk forecast shows slight upward drift, consistent with the ARIMA trend.
 - Risk: The flat forecast limits expected gains, and increasing volatility (~0.26% by period 5) could lead to fluctuations. The CI lower bound (\$236.77) suggests moderate downside risk.

9. Additional Insights

- **Model Limitations:**
 - The ARIMA models (random walk for Costco and JPMorgan, random walk with MA(1) for Walmart) are simple and reflect the efficient market hypothesis, where stock prices are hard to predict beyond trends.
 - Convergence warnings in ARIMA fitting suggest potential instability in higher-order models, but the selected models are robust due to low AICs.
 - GARCH models show weak volatility clustering for Costco but stronger effects for Walmart and JPMorgan, informing risk levels.
- **Volatility Context:**
 - Costco's low volatility (~0.15%) supports a short position by reducing the risk of large price swings.
 - Walmart's moderate volatility (~0.32%) is acceptable for a buy, as the upward trend outweighs fluctuation risks.
 - JPMorgan's increasing volatility (~0.25% to 0.26%) warrants caution but doesn't negate the bullish trend.
- **No Cointegration:** The lack of cointegration means trading decisions are based on individual stock dynamics, not relative movements.
- **Random Walk Comparison:**
 - Random walk forecasts show slight upward drift for all stocks, but they are less reliable than ARIMA due to their reliance on noise (standard deviation of differences).
 - ARIMA forecasts are preferred for trading decisions due to their statistical grounding.

10. Summary Table of Recommendations

Stock	ARIMA Model	Forecast Change	Trend	Volatility (GARCH)	Recommendation
Costco	(0,1,0)	0.00	Downward	~0.15% (low)	Short
Walmart	(0,1,1)	+0.04	Upward	~0.32% (moderate)	Buy
JPMorgan	(0,1,0)	0.00	Upward	~0.25% (moderate)	Buy

11. Caveats and Considerations

- **Short-Term Horizon:** The recommendations are for a 30-period horizon (likely 30 days, assuming daily data). Longer-term trends may differ.
- **Market Efficiency:** The simple ARIMA models (random walks) suggest limited predictability, so expected gains/losses are modest.
- **External Factors:** The analysis is purely technical. Fundamental factors (e.g., earnings, macroeconomic conditions) or market sentiment could override these signals.
- **Risk Management:** Use stop-loss orders, especially for Costco (short) and JPMorgan (buy), due to potential CI ranges and volatility.
- **Data Scale:** GARCH warnings suggest rescaling returns for better convergence, but results are still valid. Future analysis could rescale for robustness.

Conclusion

The analysis indicates a short position for Costco due to its downward trend and low volatility, and buy positions for Walmart and JPMorgan due to their upward trends and moderate volatility. These recommendations are based on ARIMA forecasts and GARCH volatility insights, with Walmart showing the strongest bullish signal (+0.04 forecast change). For real-world trading, complement this analysis with fundamental data and market context to mitigate risks.

INTERPRETACIONES PERSONALES

Empecemos con Costco. El análisis recomienda vender (hacer "short"), lo que significa apostar a que el precio va a bajar. El modelo ARIMA dice que el precio se quedará plano en \$985.92, pero hay una señal de "tendencia bajista". Esto me hace pensar que, aunque no se espere un desplome, podría haber una pequeña caída. Lo bueno es que la volatilidad (o sea, qué tan loca se pone la acción) es súper baja, como un 0.15%, así que no parece que vaya a haber subidas locas que nos arruinen la jugada. Personalmente, me da curiosidad este "bajista" porque Costco es una empresa sólida, pero si los números dicen que puede caer un poco, quizás es buena idea probar con una venta corta, siempre con cuidado. El riesgo está en que el precio podría no moverse mucho o incluso subir un poquito, según el intervalo de confianza que da el modelo.

Ahora, hablemos de Walmart. Acá la recomendación es comprar, y la verdad, me gusta cómo pinta. El modelo ARIMA predice un pequeño aumento de \$95.36 a \$95.40, y dice que la tendencia es "alcista". No es que te vayas a hacer millonario, pero ese pequeño empujón hacia arriba suena prometedor. La volatilidad es moderada (0.32%), lo que significa que el precio puede moverse un poco, pero no es una montaña rusa. Para mí, Walmart parece una apuesta segura para quienes quieren algo de ganancia sin tanto drama. La confianza del modelo (el intervalo de confianza) también dice que el precio podría subir hasta \$96.85, lo

cual refuerza la idea de que hay chance de ganar algo. El riesgo es que la ganancia no sea enorme, y si el precio se tambalea, podrías no ver mucho movimiento.

Finalmente, JPMorgan. También recomiendan comprar, pero aquí estoy un poco más dudoso. El modelo ARIMA dice que el precio se queda en \$241.60, sin cambios, aunque tiene una etiqueta de “tendencia alcista”. Esto me suena a que hay un poquito de optimismo, pero no tanto como con Walmart. La volatilidad es moderada (0.25% y subiendo), así que hay más riesgo de que el precio se mueva para cualquier lado. El intervalo de confianza dice que el precio podría llegar a \$246.43, lo cual no está mal, pero también podría bajar a \$236.77. Mi interpretación es que JPMorgan podría ser una buena compra si crees que el sector bancario va a estar fuerte, pero hay que estar atentos porque la volatilidad está subiendo. Comparado con Walmart, siento que esta es una apuesta un poco más arriesgada.

En resumen, mi visión personal es que Walmart parece la opción más atractiva para comprar, con una pequeña ganancia casi asegurada y un riesgo manejable. Costco podría ser interesante para una jugada corta, pero hay que ser cuidadosos porque el precio podría no moverse mucho. JPMorgan me genera dudas; creo que podría funcionar, pero la volatilidad me hace pensar dos veces. Eso sí, este análisis es puro números, y en la vida real, cosas como noticias, reportes de ganancias o el humor del mercado pueden cambiar todo.

El análisis usa precios de cierre diarios (107 días) y herramientas como ARIMA, GARCH y pruebas de estacionariedad. Los precios no son estacionarios, sino que siguen un “paseo aleatorio” (random walk), o sea, suben y bajan sin patrón claro. Esto me dice que predecir es difícil; el mercado es eficiente y no hay trucos fáciles.

Los correlogramas muestran que los precios de hoy se parecen a los de ayer, pero al diferenciarlos, se vuelven ruido aleatorio. No hay patrones complicados, lo que simplifica las cosas, pero también significa que no hay señales claras para ganar fácil.

El test de cointegración dice que Costco, Walmart y JPMorgan no están conectados a largo plazo. Cada acción va por su lado, así que no podemos asumir que se mueven juntas. Esto me decepciona un poco, porque no podemos hacer jugadas de pares, pero simplifica el análisis.

Los modelos ARIMA predicen: Costco se queda en \$985.92 (bajista), Walmart sube a \$95.40 (alcista), y JPMorgan se queda en \$241.60 (alcista). Walmart me emociona por su pequeño aumento; Costco parece candidato para vender; JPMorgan es un volado. El random walk da subidas ligeras, pero confío más en ARIMA.

La volatilidad (GARCH) es baja en Costco (0.15%), moderada en Walmart (0.32%) y moderada pero subiendo en JPMorgan (0.25%-0.26%). Costco es tranquilo, ideal para vender. Walmart es manejable para comprar. JPMorgan me preocupa por su volatilidad creciente.

Recomendaciones: vender Costco (bajista, baja volatilidad), comprar Walmart (subida clara, riesgo aceptable), comprar JPMorgan (alcista, pero arriesgado). Walmart es mi favorito; Costco puede dar algo con una venta corta; JPMorgan es para valientes.

